STUDY • TYPE • RUN • INTERPRET

A COMPLETE, ILLUSTRATED STUDENT MANUAL

R FROM ZERO.

Learn R and RStudio step by step, from your very first click to probability, statistics and regression.

01	02	03	04
INSTALL	WRITE CODE	ANALYZE DATA	INTERPRET

PREPARED AND AUTHORED BY

Author: Azizbek Ilkhomzoda

HOW TO USE THIS BOOK

Your learning route

This is a practical workbook, not a collection of commands. Follow each screen and do the exercises.

STARTING LEVEL: ZERO No coding or statistical software experience is assumed. Windows instructions are shown in detail; shortcuts for macOS are provided where relevant.

1. **Read the CLICK instruction**
   Each task tells you exactly which menu, button or tab to choose.
2. **Copy the R code**
   Use the Source editor in RStudio, not a Word document or web search bar.
3. **Press Run**
   Press **Ctrl + Enter** (Mac: Cmd + Enter) or the Run button.
4. **Check the result**
   Confirm your Console or Plots pane matches the example. Explain what your number means.

Where this fits in STAT201

Course part	Skills you will build
Weeks 1–2	Variables, data, mean, median, SD, tables and charts
Weeks 3–5	Probability, Bayes, binomial and normal distributions
Weeks 6–7	Covariance, correlation, sampling and the CLT
Weeks 9–11	Confidence intervals, hypothesis testing, group comparisons
Weeks 12–15	Regression, final data project, exam practice

ONE SIMPLE RULE
Never type an unfamiliar command without first asking: What question am I answering, and what does the result measure?

> All student data used here are fictional training examples. Week 8 is not scheduled in our course plan.

PAUSE & PRACTICE
Before beginning, find an empty folder where your STAT201 project will live.

☐ DONE

GETTING STARTED

Step 1 · Download R

First install the R programming language. R performs the calculations; RStudio comes later.

1. **Open a browser**
   In Chrome, Edge or Firefox, type **https://cran.r-project.org/** in the address bar and press Enter.

2. **Choose Windows**
   Click **Download R for Windows**. On the next page, click **base**.

3. **Download the installer**
   Click **Download R-[latest release] for Windows**. The version number changes over time.

ILLUSTRATED SCREEN GUIDE · INTERFACE MAY VARY

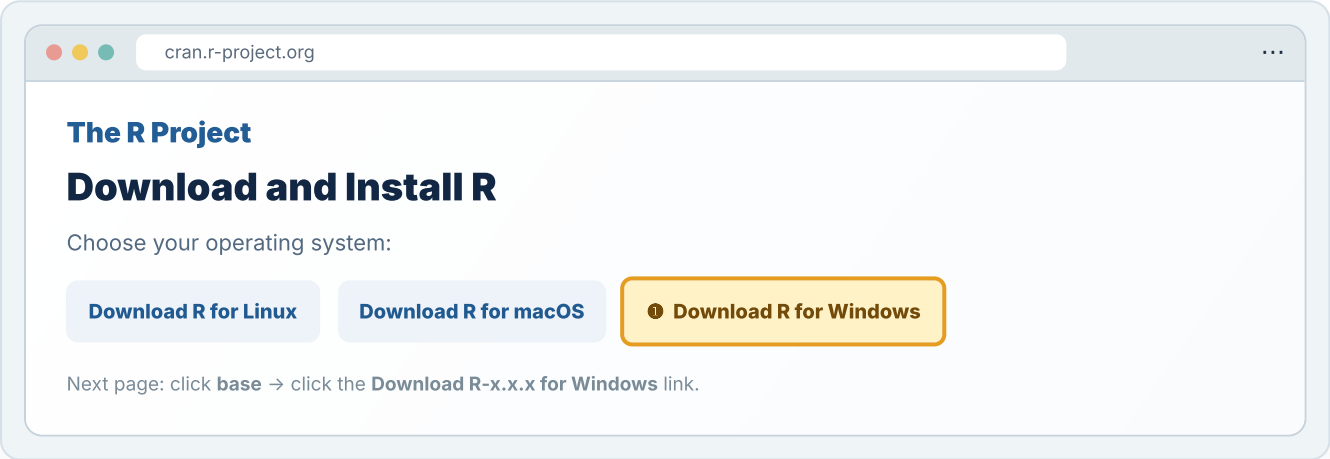


Figure 1. R download navigation. Click the highlighted Windows link; then choose base.

IMPORTANT
Download the stable release from **CRAN**. Do not choose R-devel or experimental builds.

MAC / LINUX
Choose your operating system at CRAN, install R first, then return to the next page for RStudio.

PAUSE & PRACTICE
CHECK: CRAN homepage → Download R for Windows → base → latest release.

☐ DONE

GETTING STARTED

Step 2 · Install R

After downloading R, open the installer and complete the setup wizard.

1. **Open the downloaded file**
   Open your Downloads folder and double-click the R installer ending in **.exe**.

2. **Follow the setup wizard**
   Choose a language, click **Next** or **OK**, and leave the default installation folder and typical components selected.

3. **Finish**
   Choose **Install** / **Next** as prompted, then **Finish**. R is now installed on your computer.

ILLUSTRATED SCREEN GUIDE · INTERFACE MAY VARY

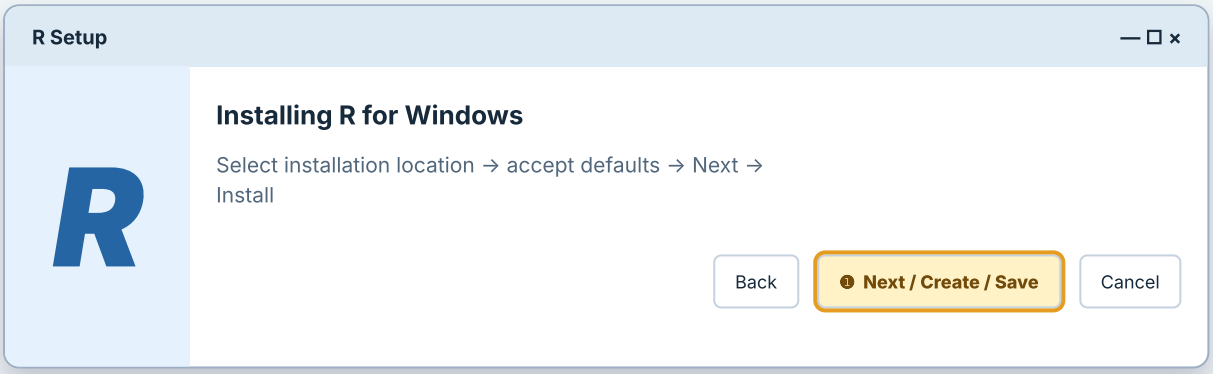


Figure 2. Typical Windows installation wizard. The button names and order may vary by version.

CHECKPOINT
R is the calculation engine, but its basic interface is not necessary for this course. Next, install **RStudio Desktop**, which is easier for beginners.

COMMON MISTAKE
R and RStudio are **two separate programs**. Installing RStudio alone does not replace R.

PAUSE & PRACTICE
CHECK: The R installer has finished; you can now begin RStudio installation.

☐ DONE

GETTING STARTED

Step 3 · Install RStudio

RStudio is where you will write, run, save and inspect your statistical work.

1. **Visit the official website**
   Open **https://posit.co/downloads/** in your browser.

2. **Choose the free edition**
   Find **RStudio Desktop – Open Source / Free** and click **Download**.

3. **Install RStudio**
   Open its installer, accept standard options and finish. Then launch **RStudio** from the Start menu.

ILLUSTRATED SCREEN GUIDE · INTERFACE MAY VARY

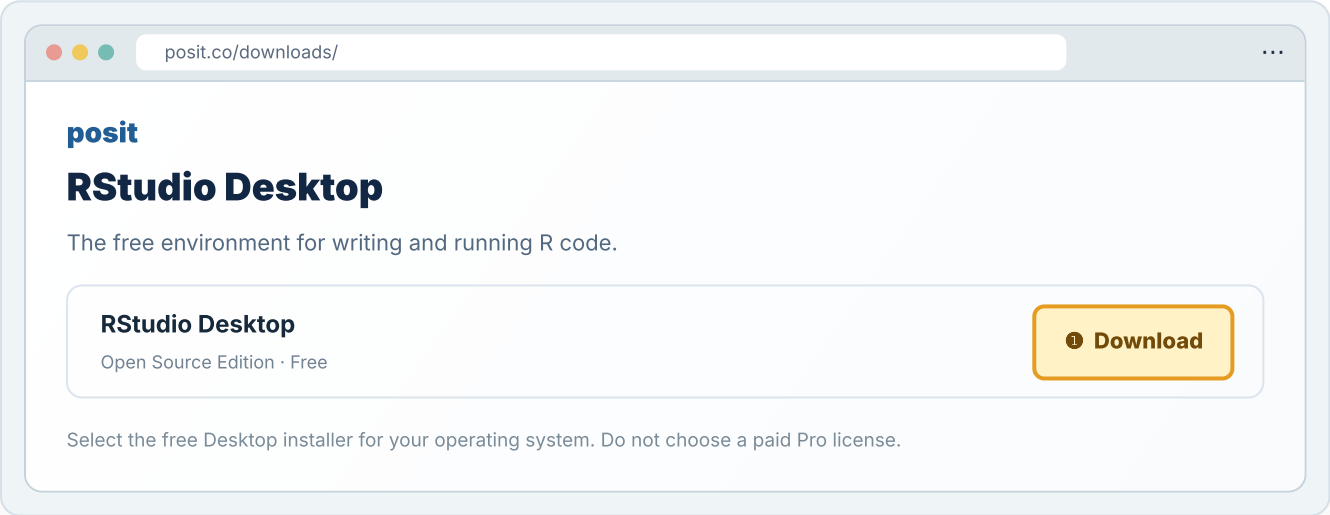


Figure 3. Choose the free RStudio Desktop download.

TEST IT
When RStudio opens, locate the **Console**; type `1 + 1` and press **Enter**. The output should end in `2`.

VERSIONS
This guide follows the modern RStudio Desktop layout. Your screen may look slightly different depending on the version.

PAUSE & PRACTICE
CHECK: R installed first, RStudio Desktop second; Console outputs [1] 2.

☐ DONE

THE INTERFACE

Step 4 · Meet RStudio

Four work areas. The most important distinction: write in Source, read results in Console.

ILLUSTRATED SCREEN GUIDE · INTERFACE MAY VARY

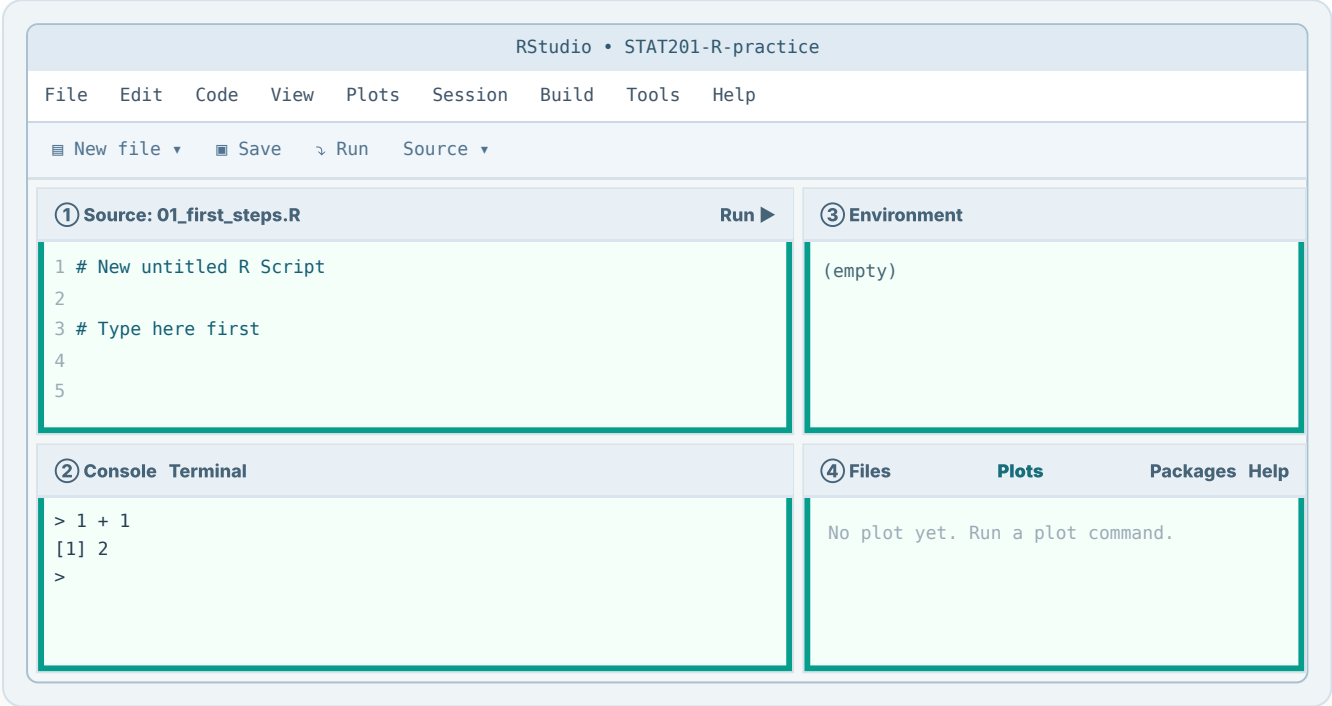


Figure 4. Four principal RStudio panes: Source, Console, Environment and Output.

① SOURCE
Write and save code in an .R file.

② CONSOLE
See executed code and numerical results.

③ ENVIRONMENT
See your named objects and datasets.

④ OUTPUT
Files, Plots, Packages and Help tabs.

DO THIS NOW

Click in the Console, type `1 + 1` and press Enter. You should see `[1] 2`. `[1]` is an output-position label, not part of the answer. If Source is missing, open a new R Script in Step 6.

PAUSE & PRACTICE

Point at the four panes and say what each is for without looking at the labels.

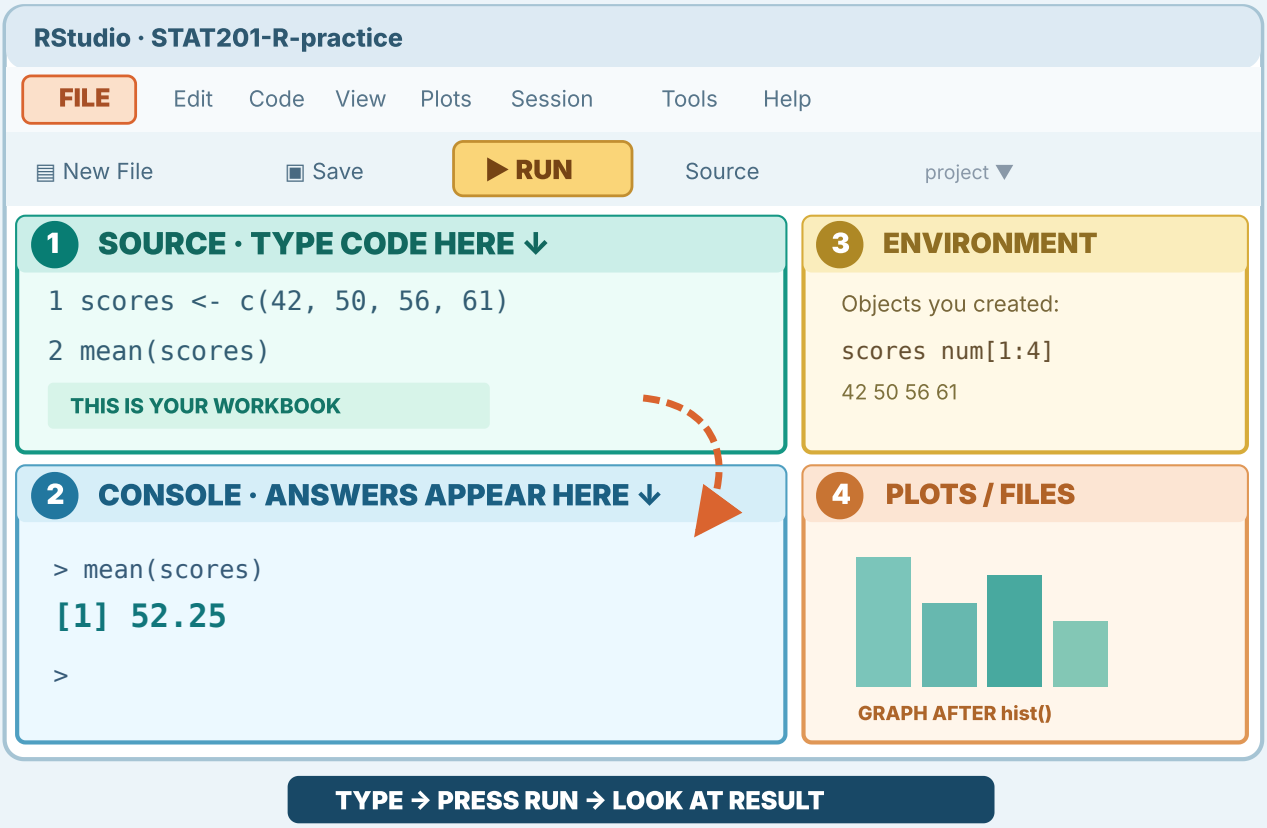
☐ DONE

ESSENTIAL MAP

Meet the WHOLE RStudio window

Think of RStudio as four workspaces inside one big window. You will see this map throughout the handbook.

1 → 2 **Keep your eyes on the numbers.**
Write in box ①, press the Run button, then look in boxes ②, ③ or ④ to see what changed.



Original teaching diagram. The four areas always have the same jobs. Pane positions can be moved in RStudio settings.

A **FILE menu**
Start a project, open a new R script, or save a file.

B **RUN ▶ button**
Runs the selected line from Source. Shortcut: Ctrl + Enter.

1 **SOURCE (top left)**
The notebook: you type and save R code here.

2 **CONSOLE (bottom left)**
R writes numeric answers and errors here.

3 **ENVIRONMENT (top right)**
Variables and tables you created appear here.

4 **PLOTS / FILES (bottom right)**
Graphs appear in Plots; saved files appear in Files.

ESSENTIAL MAP

Where to click: File · Script · Run

Follow the orange markers from left to right. You only need these controls to begin.

1 MAKE A SCRIPT

File menu ↓

```
File → New File
→ R Script
```

This creates the blank page in Source.

2 TYPE & SAVE

Top-left box ↓

```
scores <- c(4,6,8)
mean(scores)
```

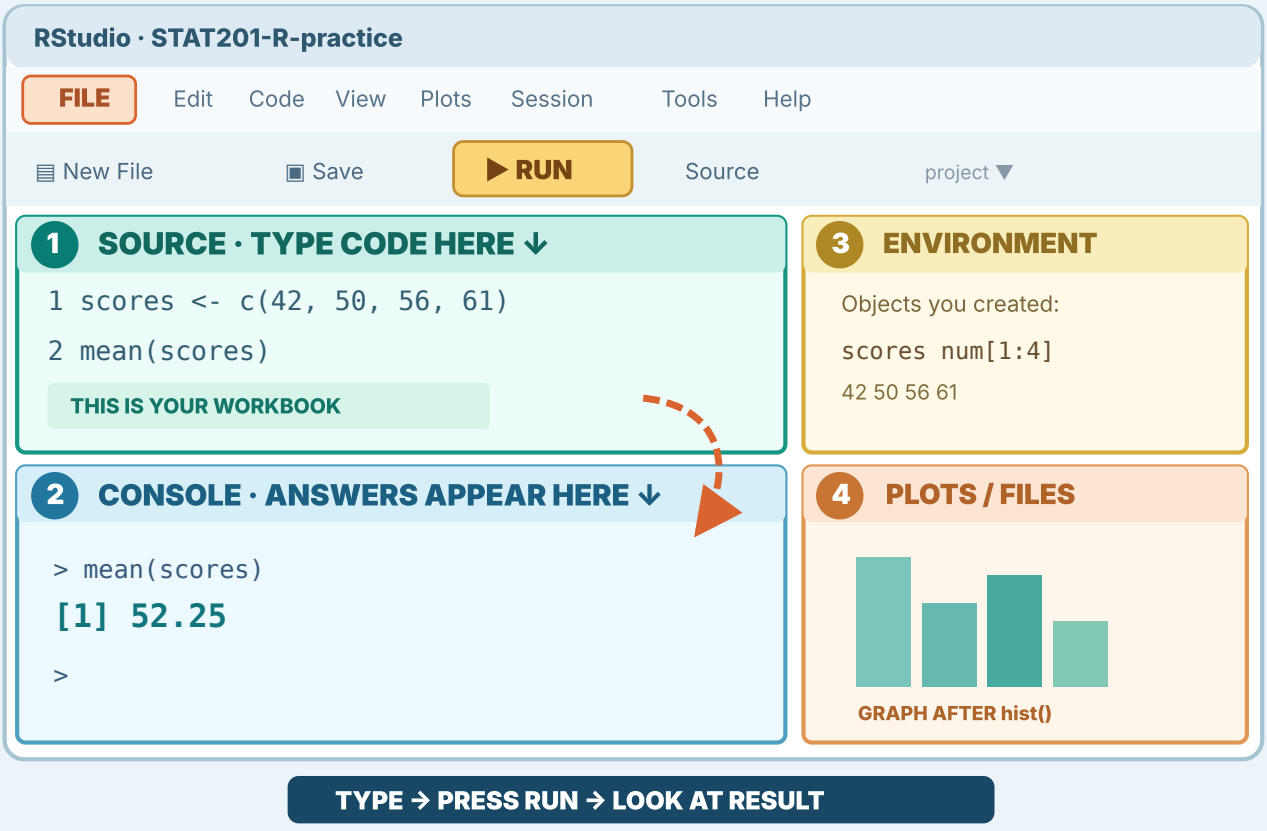
Type exactly, then press Ctrl + S to save the file.

3 RUN & SEE

Run ▶ button

```
Ctrl + Enter
Console: [1] 6
```

Press Run; see your number in the Console below.



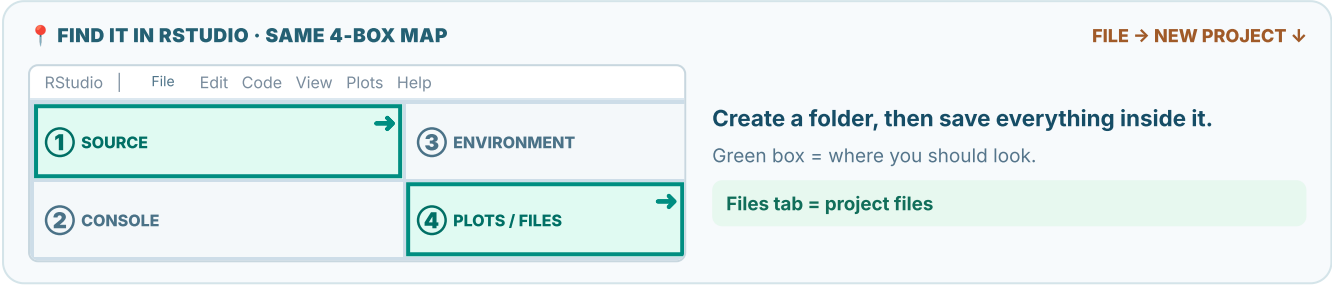
Original teaching diagram. The four areas always have the same jobs. Pane positions can be moved in RStudio settings.

✓ CHECK IT Show a classmate the **File** menu, **Source**, **Run**, and **Console** without looking at this page.

THE INTERFACE

Step 5 · Create a project

A project is a folder that stores your scripts, datasets and figures together.



1 **Open the project wizard**
In RStudio, click **File → New Project...**.

2 **Choose how to create it**
Click **New Directory → New Project**.

3 **Name and create**
Type **STAT201-R-practice**, choose a parent folder you can find again, then click **Create Project**.

ILLUSTRATED SCREEN GUIDE · INTERFACE MAY VARY

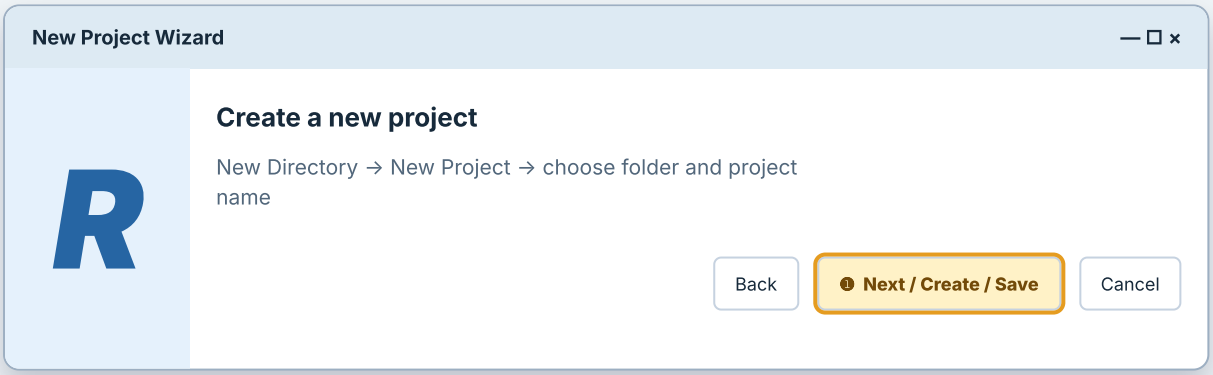


Figure 5. Project wizard (simplified). Work inside this folder in every lesson.

📁 **STAT201-R-practice**
Your one project folder

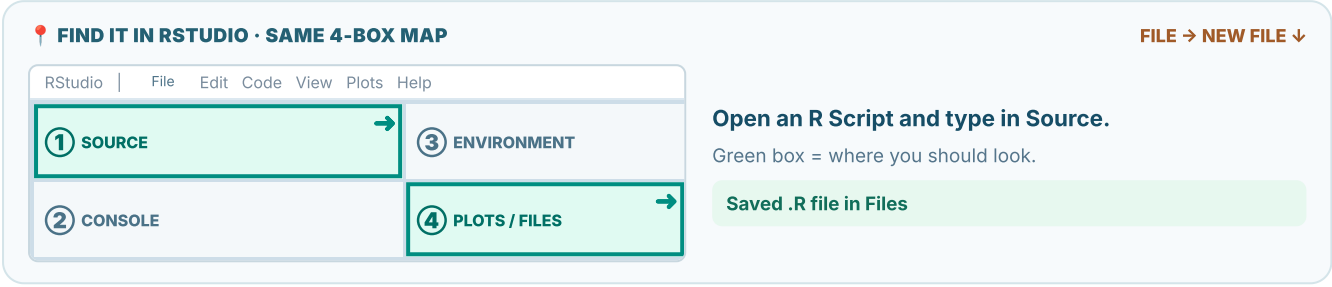
```
├── 01_first_steps.R
├── students.csv
└── graphs and saved results
```

WHY THIS MATTERS
Files such as `students.csv` are easiest to import when they are kept inside the current RStudio project. Avoid scattering files across Downloads, Desktop and unrelated folders.

THE INTERFACE

Step 6 · Open your first R script

The Source editor is the correct place for reusable statistical work.



1 **Open a new script**
Click **File → New File → R Script** (or press **Ctrl + Shift + N** on Windows).

2 **Write one line**
Click in the blank top-left editor and type `2 + 3`.

3 **Save it**
Press **Ctrl + S**. Name the file **01_first_steps.R** and save it inside your project.

ILLUSTRATED SCREEN GUIDE · INTERFACE MAY VARY

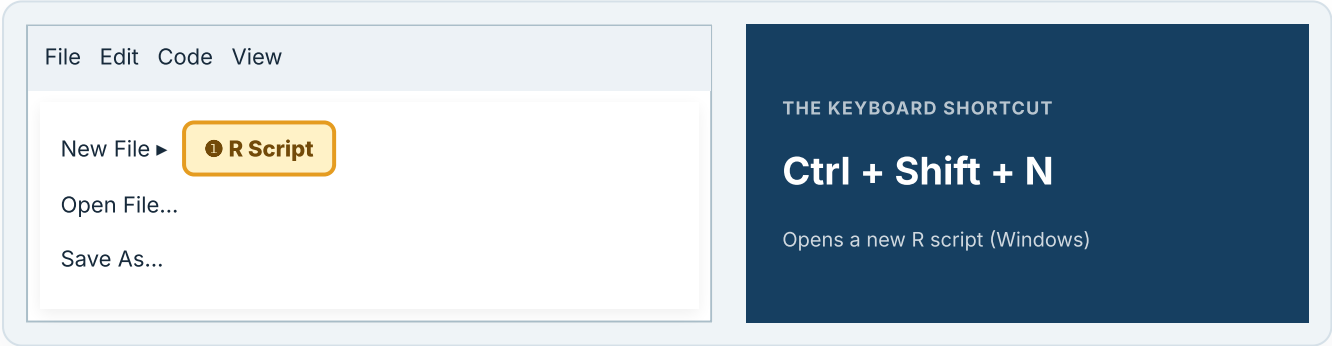


Figure 6. Find the R Script command under the File menu.

TYPE THIS IN THE SOURCE EDITOR

```
2 + 3
10 / 2
2 ^ 3
sqrt(49)
# This is a comment: R will not run this line.
```

WHAT YOU SHOULD SEE `2 + 3 -> 5; 10 / 2 -> 5; 2 ^ 3 -> 8; sqrt(49) -> 7`

THE INTERFACE

Step 7 · Press Run

Writing and running code are two different actions. The Run button sends code to R.

FIRST TYPE BOTH LINES IN THE SOURCE EDITOR

```
scores <- c(42, 50, 56, 61)
mean(scores)
```

LOOK AT GREEN PANELS: SOURCE → RUN → CONSOLE

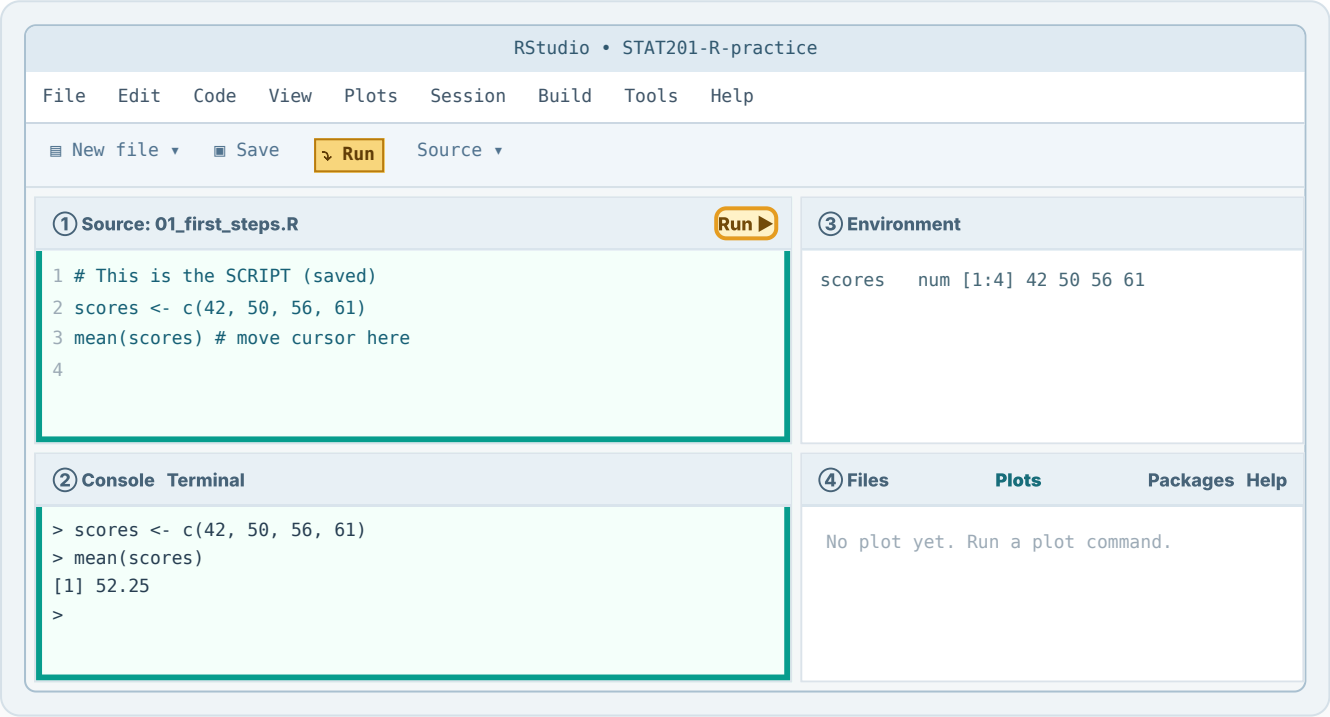


Figure 7. Put the cursor on a line in Source, then press Run (or Ctrl + Enter).

1. **Click the Source editor**
   Place the blinking cursor anywhere on the line `mean(scores)`.

2. **Execute that line**
   Press **Ctrl + Enter** (Mac: **Cmd + Enter**) or click the editor's **Run** ▶ button.

3. **Find the output**
   Look in the bottom-left **Console** and identify the printed numerical result.

SOURCE = SAVED
Keep your commands in the **.R** script so you can reproduce the calculation.

CONSOLE = OUTPUT
The Console executes lines and shows results. It is not your final assignment file.

MINI LABORATORY

Watch the window CHANGE in real time

These three tiny tasks show what changes in RStudio when a command runs. Do them in this order.

① → ③ **The rule:**
You always type on the left. Depending on your code, a number, a variable or a graph appears somewhere else.

A MAKE DATA

→ ③ Environment

```
scores <- c(4, 6, 8)
```

A new object named scores appears in the top-right box.

B FIND THE MEAN

→ ② Console

```
mean(scores)
[1] 6
```

A numeric answer 6 appears in the bottom-left box.

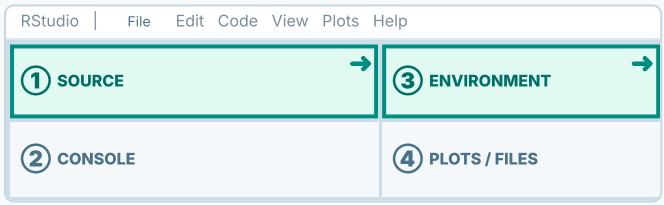
C DRAW A GRAPH

→ ④ Plots

```
hist(scores)
```

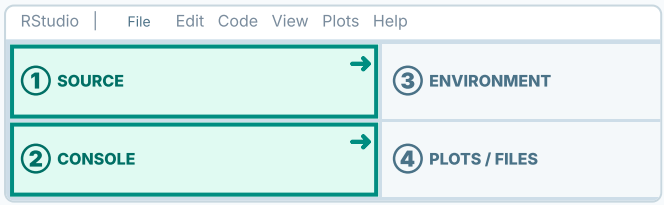
A histogram appears under the bottom-right Plots tab.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP — TASK A: MAKE AN OBJECT ↓



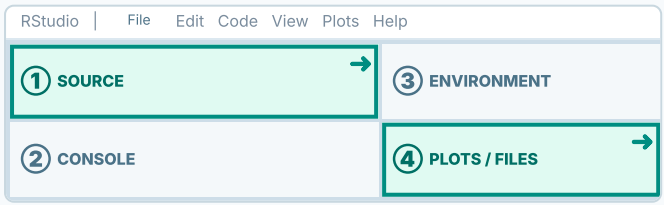
Write the scores vector.
Green box = where you should look.
Top right: scores appears

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP — TASK B: CALCULATE ↓



Write mean(scores).
Green box = where you should look.
Bottom left: [1] 6

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP — TASK C: DRAW ↓



Write hist(scores).
Green box = where you should look.
Bottom right: histogram

Remember: The Console does not "turn into" the plot. RStudio has different areas for different kinds of results.

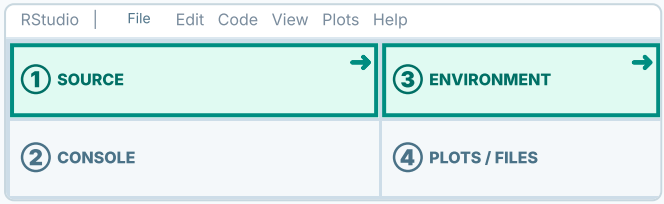
STATISTICS FOUNDATIONS

What is ONE variable?

Before you calculate anything, understand the difference between an observation, variable and object.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

ONE VARIABLE ↓



A single column such as Score.

Green box = where you should look.

Named vectors appear in Environment

REAL QUESTION

"Do students with more study hours tend to get higher exam scores?"

Student (row)	Hours (variable 1)	Score (variable 2)	Group (variable 3)
1	1	42	A
2	2	50	B
3	3	56	A
4	3	61	B

ONE OBSERVATION

One student = one row.

ONE VARIABLE

One measured characteristic = one column.

UNIVARIATE

Study score alone: its mean, median, histogram.

BIVARIATE

Study score versus hours: correlation, regression.

TYPE THIS IN THE SOURCE EDITOR

```
age <- 19                  # one R object holding one value
scores <- c(42, 50, 56, 61)  # one object holding four values
length(scores)              # 4 observations
```

BE CAREFUL

A **statistical variable** is a characteristic measured for many units. An **R object** is a named storage location. A vector named `scores` may store all values of a statistical variable.

STATISTICS IN PICTURES

Variable = COLUMN · Observation = ROW

This is what a variable means in statistics. Learn this picture before you begin calculating.

THE WHOLE DATASET · 3 VARIABLES

Student	Hours	Score
1	2	50
2	4	67
3	5	74

GREEN COLUMN = one variable: Hours. Three recorded values of that variable are 2, 4, 5.

ZOOM IN ON ONE ROW →

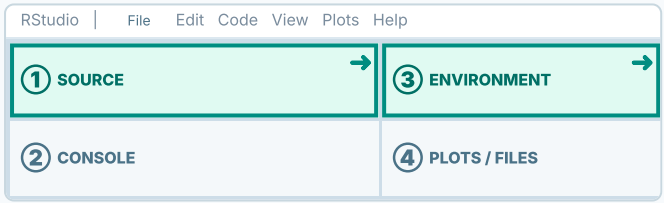
Student	Hours	Score
2	4	67

GREEN ROW = one observation: student 2 studied 4 hours and scored 67.

In the TABLE → **In R:** `data.frame(...)` → **See it in Environment**

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

CREATE DATA ↓



Type code in Source.

Green box = where you should look.

Dataset name appears in Environment

ONE VARIABLE

score

One characteristic measured for all students. A column.

ONE OBSERVATION

Student 2

One student with several recorded characteristics. A row.

ONE R OBJECT

scores ← ...

A named object in R which stores the column values.

We do **one-variable statistics** when we analyze score alone (mean, histogram). We do **two-variable statistics** when comparing hours with score (correlation, regression).

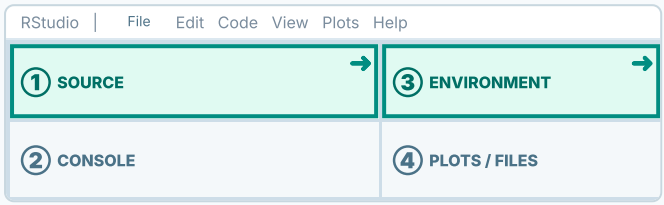
YOUR FIRST DATA

Write values and understand symbols

R is a calculator and a programming language. These few symbols are the building blocks.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

TYPE + CREATE ↓



Use ← to make a named object.

Green box = where you should look.

Object name appears top right

TYPE THIS IN THE SOURCE EDITOR

```
x <- 8                # assign 8 to object x
x                      # print x
a <- c(3, 5, 7, 9)    # c() combines values into a vector
mean(a)                # arithmetic mean = 6
sum(a)                 # sum = 24
a[2]                   # second element = 5
a[a >= 7]              # select values 7 and 9
length(a)              # number of values = 4
```

Symbol	Meaning	Example
`<-`	Store/assign	`x <- 8`
`c()`	Combine values	`c(3, 5)`
`#`	Comment	`# mean`
`==`	Compare for equality	`x == 8`
`$`	Select dataset column	`students$score`
`[]`	Select items	`a[2]`
`NA`	Missing value	`c(7, NA)`

? YOUR TURN

Create `prices <- c(10, 20, 30)`. Before running `mean(prices)`, calculate the expected result by hand. Answer: **20**.

YOUR FIRST DATA

Build a dataset of 10 students

This is the dataset we use throughout the book; no separate download is required.

ONE ROW = ONE STUDENT

Each vector below has ten entries, in matching order. We deliberately use a small fictional class so the results are easy to check.

TYPE THIS IN THE SOURCE EDITOR

```
students <- data.frame(
  id    = 1:10,
  hours = c(1, 2, 3, 3, 4, 5, 5, 6, 7, 8),
  score = c(42, 50, 56, 61, 63, 68, 73, 78, 85, 91),
  group = c("A", "B", "A", "B", "A",
            "B", "A", "B", "A", "B")
)

head(students)    # view first six rows in Console
View(students)    # open a spreadsheet-style viewer
str(students)     # check data types
nrow(students)    # 10 rows
```

LOOK AT GREEN PANELS: SOURCE → ENVIRONMENT

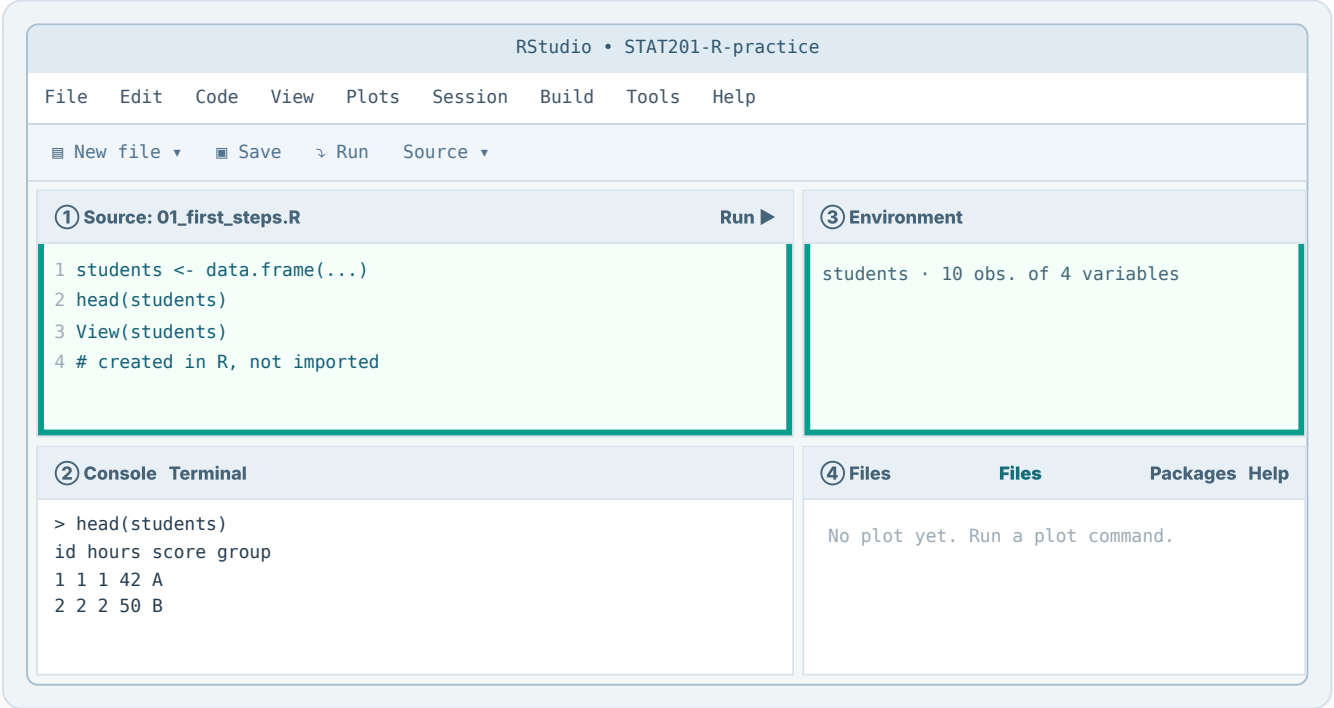


Figure 8. After running the code, RStudio lists students in Environment.

THE $ SIGN

`students$score` means: take the **score column** from the table called students. The `id` column is an identifier, not a measurement you should average.

YOUR FIRST DATA

Save a CSV and open it again

CSV means comma-separated values. Saving a dataset helps you practice real file import.

1. **Create the file**
   Run the first command below; it writes **students.csv** inside your RStudio project.
2. **Find it**
   Click the bottom-right **Files** tab. You should see **students.csv** beside your script.
3. **Read the file**
   Run `read.csv(...)`, then use `View(...)` to inspect it.

TYPE THIS IN THE SOURCE EDITOR

```
write.csv(students, "students.csv", row.names = FALSE)

loaded <- read.csv("students.csv")
head(loaded)
View(loaded)
nrow(loaded)      # should be 10
names(loaded)     # id, hours, score, group
```

LOOK AT GREEN PANELS: SOURCE → FILES / ENVIRONMENT

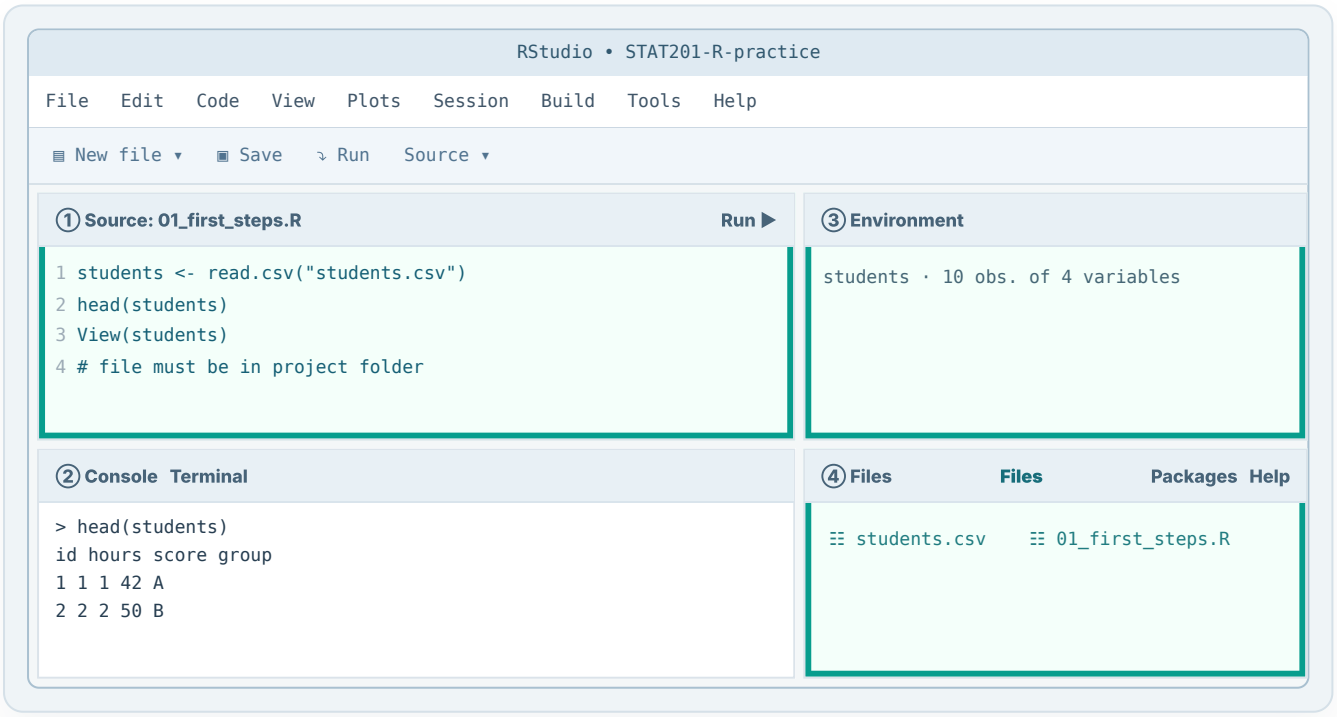


Figure 9. The Files tab is in the bottom-right RStudio pane.

FILE NOT FOUND?

Run `getwd()` to see your project folder and `list.files()` to see filenames. Keep the CSV in that folder; do not type the filename differently.

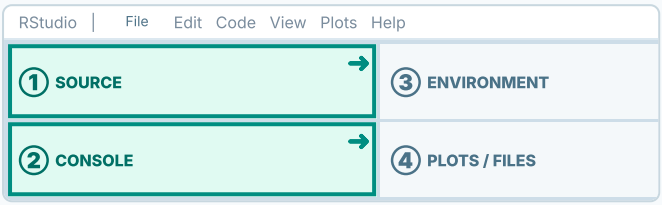
STATISTICS · WEEK 1

Compute mean and median

Descriptive statistics summarize one variable. Always state the unit of measurement.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

CALCULATE ↓



Run mean() and median().

Green box = where you should look.

Read numbers in the Console

TYPE THIS IN THE SOURCE EDITOR

```
x <- students$score
mean(x)       # arithmetic mean
median(x)     # middle of the ordered sample
min(x)        # smallest value
max(x)        # largest value
summary(x)    # summary with quartiles
```

66.7	65.5	10
MEAN (POINTS)	MEDIAN (POINTS)	SAMPLE SIZE

Question	What to report
What is the typical score?	Mean = 66.7 points; median = 65.5 points.
Why report both?	The mean uses all values; the median is less sensitive to extreme values.
What must not be claimed?	These 10 fictional students do not prove anything about all TIU students.

Sample mean: $\bar{x} = (x_1 + x_2 + \cdots + x_n) / n$

QUICK CHECK

Change the last score from 91 to 191 (an extreme, unrealistic value) and compare the new **mean** with the **median**. Which one changes more?

PAUSE & PRACTICE

Calculate the mean of the first four scores by hand. Compare with mean(x[1:4]).

☐ DONE

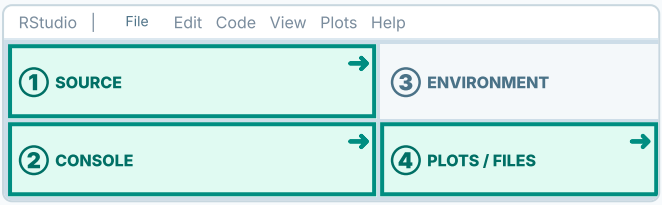
STATISTICS · WEEK 1

Variance, standard deviation and outliers

A measure of center alone cannot tell you how spread out the data are.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

MEASURE SPREAD ↓



Run sd() / var() / boxplot().

Green box = where you should look.

Numbers left, box plot right

TYPE THIS IN THE SOURCE EDITOR

```
x <- students$score
var(x)            # sample variance (uses n - 1)
sd(x)             # sample standard deviation
range(x)          # returns minimum and maximum
diff(range(x))    # range as one number
quantile(x)       # quartiles
IQR(x)            # Q3 - Q1
boxplot(x, horizontal = TRUE, col = "lightblue")
```

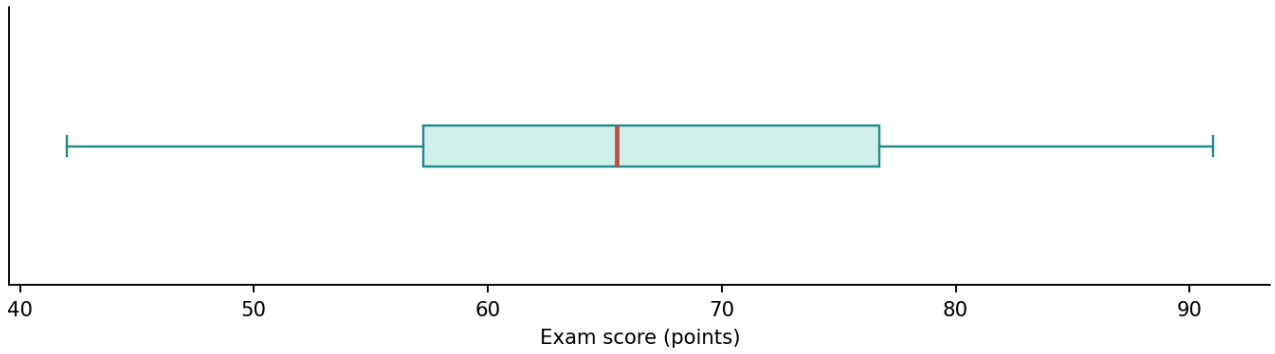


Figure 10. A boxplot summarizes the distribution and helps flag unusual values.

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1) \cdot s = \sqrt{s^2}$$

RESULT

The sample SD here is about **15.43 points**; the sample variance is about **238.23 points²**.

INTERPRETATION

Higher SD means greater variation around the mean, in the same original units as the variable.

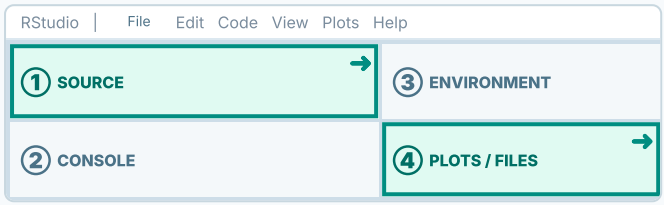
STATISTICS · WEEK 2

Make your first charts

Graphs are statistical answers. RStudio displays them in the bottom-right Plots tab.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

DRAW GRAPHS ↓



Run hist() or plot().

Green box = where you should look.

Graph goes to Plots, bottom right

1. **Choose the right display**
   A numerical variable → histogram; a categorical variable → bar chart; two numerical variables → scatterplot.

2. **Run the command**
   Put the cursor on a plotting command in Source, then press **Ctrl + Enter**.

3. **View and export**
   Click **Plots** at bottom right. Use **Export** if you need a saved picture.

TYPE THIS IN THE SOURCE EDITOR

```
hist(students$score, col = "lightblue",
     main = "Exam scores", xlab = "Score")

table(students$group)
barplot(table(students$group), col = "skyblue")

plot(students$hours, students$score,
     xlab = "Study hours", ylab = "Exam score")
```

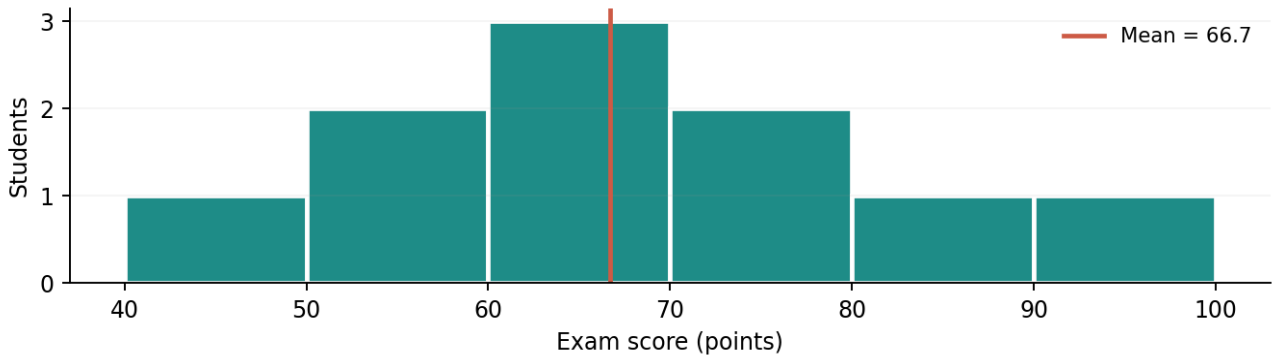


Figure 11. A histogram shows how many scores fall into each interval.

> Do not choose a line chart for unordered student records. For a histogram, bars touch because the horizontal variable is quantitative.

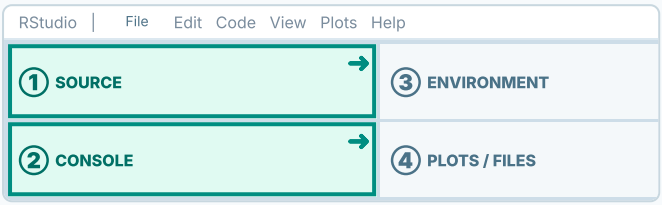
PROBABILITY · WEEK 3

Probability begins with events

Probability is a number from 0 to 1: 0 means impossible; 1 means certain.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

FIND PROBABILITY ↓



Run mean(pass) / sum(pass).

Green box = where you should look.

Probability printed bottom left

THOUGHT EXPERIMENT

Choose one of our 10 fictional students at random. What is P(pass)?

TYPE THIS IN THE SOURCE EDITOR

```
pass <- students$score >= 60     # TRUE or FALSE
pass                              # inspect values
mean(pass)                        # proportion passing
sum(pass)                         # count passing
mean(!pass)                       # proportion failing

# Basic event rules:
pA <- 0.40
pB <- 0.30
pA_and_B <- 0.10
pA_or_B <- pA + pB - pA_and_B
```

7 / 10	0.70	0.30
STUDENTS PASS	ESTIMATED P(PASS)	P(FAIL)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad | \quad P(A^c) = 1 - P(A)$$

WHY MEAN(TRUE/FALSE) WORKS

For a logical vector, R treats `TRUE` as 1 and `FALSE` as 0 when calculating the mean. This gives the sample proportion.

PAUSE & PRACTICE

What is P(fail) if 7 out of 10 pass? Use the complement rule.

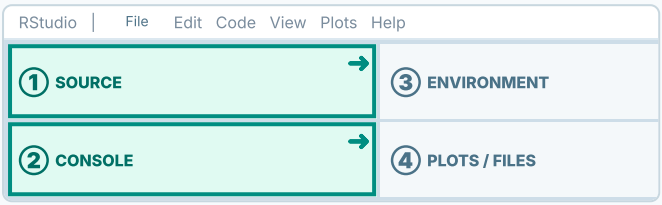
☐ DONE

PROBABILITY · WEEK 3

Conditional probability and Bayes

Conditional means “given that we already know something.”

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP



BAYES CALCULATION ↓

Run the probability formulas.

Green box = where you should look.

Result printed in Console

$$P(A \mid B) = P(A \cap B) / P(B) \quad \text{(when } P(B) > 0\text{)}$$

TYPE THIS IN THE SOURCE EDITOR

```
# A medical test example with made-up values
prevalence <- 0.02       # P(D)
sensitivity <- 0.90      # P(+ | D)
false_positive <- 0.05   # P(+ | not D)

p_positive <- sensitivity * prevalence +
              false_positive * (1 - prevalence)
p_disease_given_positive <-
  sensitivity * prevalence / p_positive

p_positive
p_disease_given_positive
```

0.067	0.269	≈27%
PROBABILITY TEST IS POSITIVE	P(DISEASE | POSITIVE)	AFTER A POSITIVE TEST

$$\text{Bayes: } P(D \mid +) = P(+ \mid D)P(D) / P(+)$$

INTERPRET CORRECTLY

Sensitivity (90%) is **not** the probability of having the disease after testing positive (about 26.9% in this constructed example). Base rates matter.

PAUSE & PRACTICE

Why is P(disease | positive) much less than test sensitivity?

☐ DONE

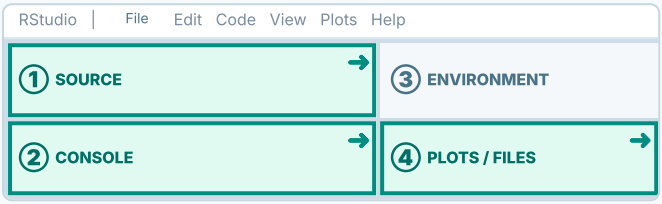
DISTRIBUTIONS · WEEK 4

Bernoulli and binomial in R

A Bernoulli trial has success/failure outcomes. The binomial counts successes in n independent trials.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

BINOMIAL ↓



Run dbinom(); barplot().

Green box = where you should look.

Number left; graph right

TYPE THIS IN THE SOURCE EDITOR

```
# X ~ Binomial(n = 5, p = 0.4)
dbinom(3, size = 5, prob = 0.4)   # P(X = 3)
pbinom(2, size = 5, prob = 0.4)   # P(X <= 2)
1 - pbinom(2, size = 5, prob = 0.4) # P(X >= 3)
qbinom(0.5, size = 5, prob = 0.4) # median quantile

set.seed(123)
rbinom(10, size = 5, prob = 0.4)  # 10 simulations
barplot(dbinom(0:5, 5, 0.4), names.arg = 0:5)
```

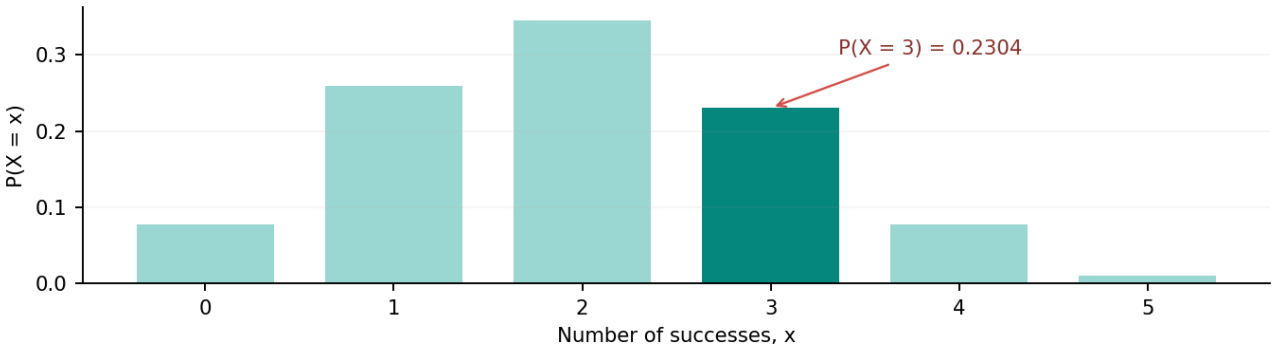


Figure 12. Probability mass function for a binomial count.

E[X] = np = 2 · Var(X) = np(1 – p) = 1.2

FOUR PREFIXES

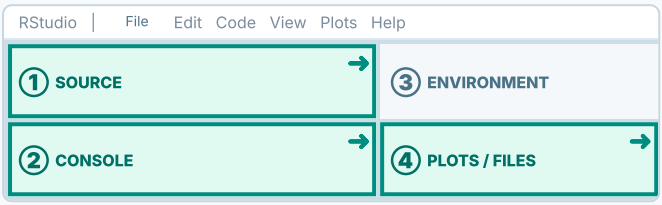
d = mass/density, **p** = cumulative probability, **q** = quantile, **r** = random simulation. These prefixes also work for normal and Poisson distributions.

DISTRIBUTIONS · WEEK 5

Normal distribution and z-scores

For a continuous variable, probability is area under the density curve—not height at one point.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP NORMAL ↓



Use pnorm() for area.

Green box = where you should look.

Probability shown in Console

TYPE THIS IN THE SOURCE EDITOR

```
# Example: X ~ Normal(mean = 70, sd = 10)
pnorm(80, mean = 70, sd = 10)    # P(X <= 80)
1 - pnorm(80, mean = 70, sd = 10)# P(X > 80)
qnorm(0.95, mean = 70, sd = 10) # 95th percentile
dnorm(70, mean = 70, sd = 10)    # density, not P(X=70)

z <- (80 - 70) / 10            # z-score = 1
set.seed(123)
rnorm(5, mean = 70, sd = 10)    # random values
```

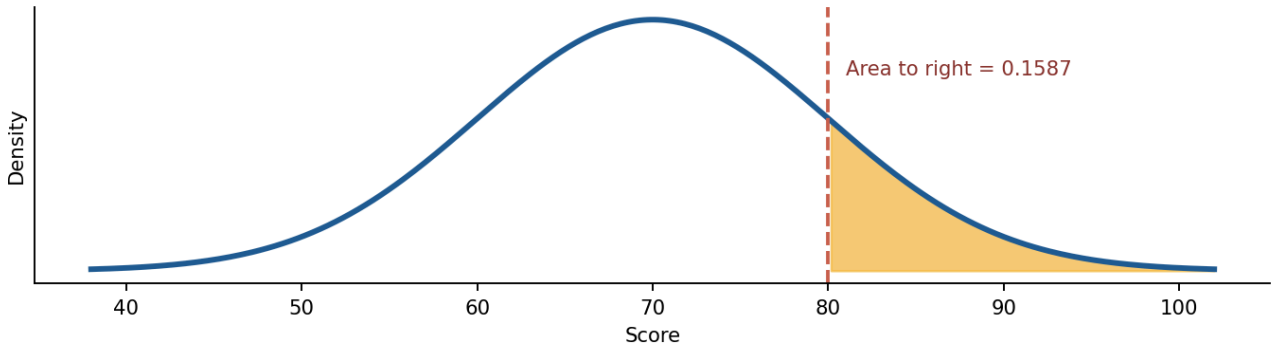


Figure 13. Yellow right-tail area is P(X > 80) ≈ 0.1587.

$z = (x - \mu) / \sigma$ · $P(X = 80) = 0$ for a continuous normal variable

COMMON CONFUSION

`pnorm()` answers probability questions about areas. `dnorm()` returns the height of the density, not the probability of one exact value.

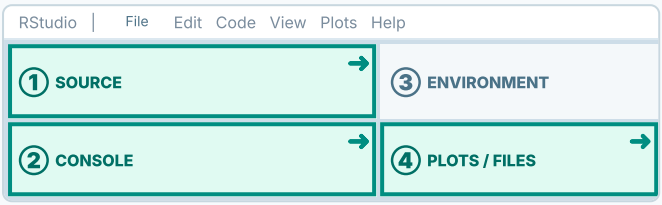
TWO VARIABLES · WEEK 6

Covariance and correlation

Are two measurements related? First draw a scatterplot, then calculate a summary.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

TWO VARIABLES ↓



Run cor() then plot().

Green box = where you should look.

Number left; scatterplot right

TYPE THIS IN THE SOURCE EDITOR

```
plot(students$hours, students$score,
     xlab = "Study hours", ylab = "Exam score")
cov(students$hours, students$score)
cor(students$hours, students$score)

# Negative association example
a <- 1:5
b <- c(10, 8, 6, 4, 2)
cor(a, b)   # -1 (perfect negative line)
```

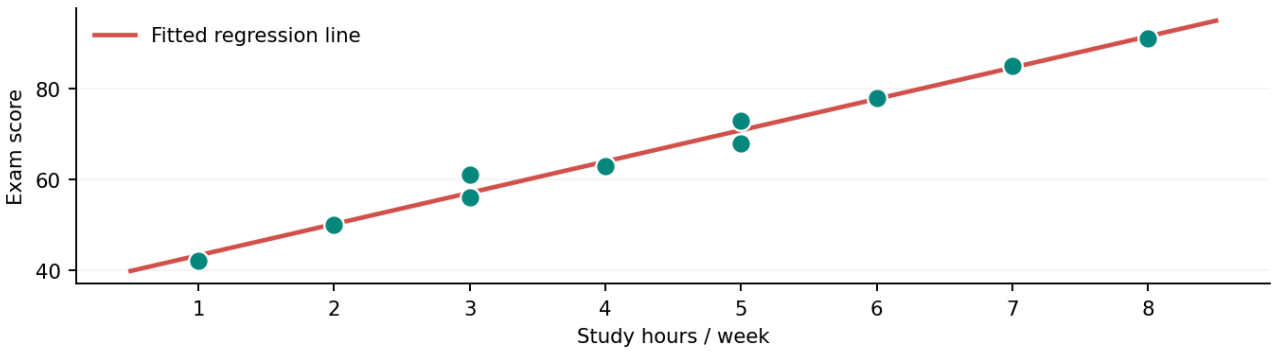


Figure 14. Study time and score are strongly positively associated in this example.

Correlation r ∈ [−1, 1]. Positive: move together. Negative: move in opposite directions.

IMPORTANT

For this fictional dataset, `cor(...)` is about **0.992**. This does **not** prove study hours cause higher scores: selection, ability or other factors may be involved.

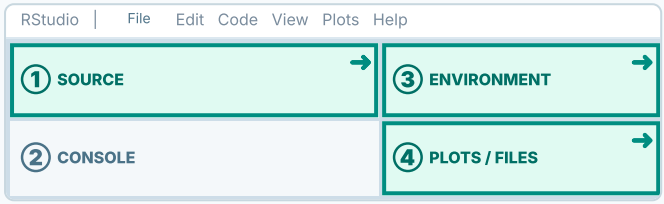
SAMPLING · WEEK 7

The Central Limit Theorem

Repeated sample means and proportions can have approximately normal distributions even when the underlying observations are not normal.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

SIMULATE ↓



Create 1000 sample proportions.

Green box = where you should look.

Object top right; histogram bottom right

TYPE THIS IN THE SOURCE EDITOR

```
set.seed(42)              # repeatable simulation
# 1000 samples, each containing 50 Bernoulli trials
proportions <- replicate(1000, {
  trials <- rbinom(50, size = 1, prob = 0.4)
  mean(trials)
})

hist(proportions, col = "lightblue",
     main = "Sampling distribution")
mean(proportions)          # close to 0.4
sd(proportions)            # close to sqrt(0.4*0.6/50)
```

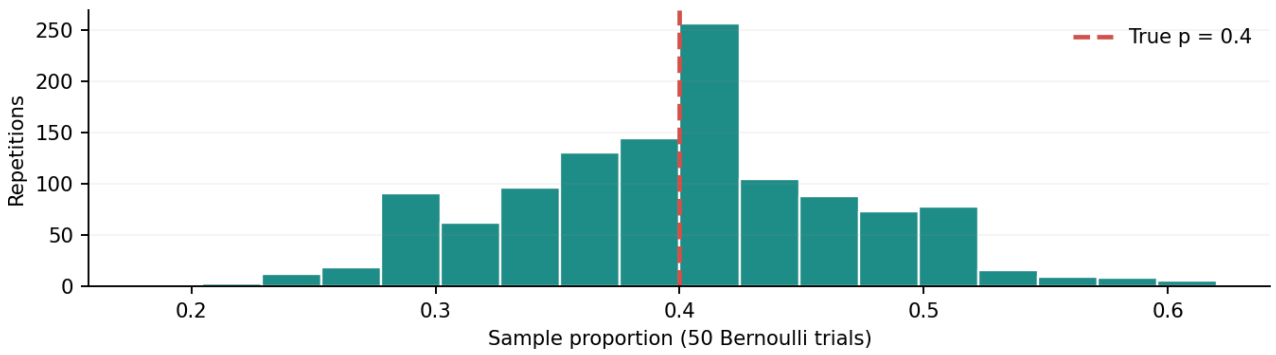


Figure 15. Distribution of many sample proportions from repeated samples (simulation).

$$SE(\hat{p}) = \sqrt{[p(1 - p)/n]} \quad \text{(for independent Bernoulli sampling)}$$

THIS IS NOT THE ORIGINAL DATA

Each bar represents **sample proportions** from repeated samples, not individual student scores. The central limit approximation requires suitable sampling assumptions.

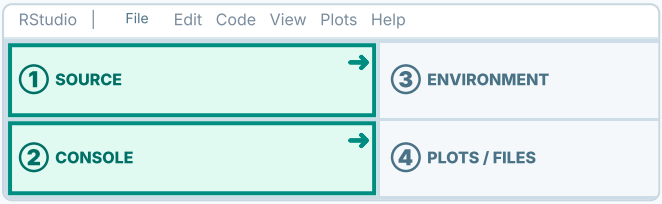
INFERENCE · WEEK 9

A confidence interval

A sample mean is an estimate. An interval shows a range of parameter values compatible with a stated procedure.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

CONFIDENCE INTERVAL ↓



Run t.test().

Green box = where you should look.

Interval printed in Console

TYPE THIS IN THE SOURCE EDITOR

```
x <- students$score
t.test(x, conf.level = 0.95)

# Understand the ingredients by hand
n <- length(x)
xbar <- mean(x)
SE <- sd(x) / sqrt(n)
critical <- qt(0.975, df = n - 1)
ci <- xbar + c(-1, 1) * critical * SE
ci
```

95% t-interval = $\bar{x} \pm t^* \times s/\sqrt{n}$

1. **Run t.test**
   Select the command and press **Ctrl + Enter**.
2. **Find the interval**
   Look for the two numbers under **95 percent confidence interval** in the Console.
3. **Interpret the method**
   Under repeated sampling, 95% of intervals built by this procedure would contain the true mean (under the assumptions).

DO NOT SAY

"There is a 95% chance that the fixed population mean lies in this particular calculated interval." That is not the standard frequentist interpretation.

PAUSE & PRACTICE

What is the standard error of the sample mean and how is it calculated?

☐ DONE

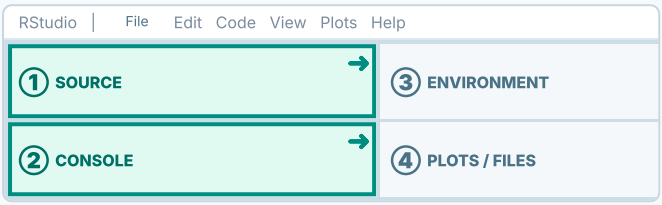
INFERENCE · WEEKS 10–11

Hypothesis tests: mean and proportion

Hypothesis tests compare observed evidence against a model under a null hypothesis.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

TEST HYPOTHESES ↓



Run t.test() or binom.test().

Green box = where you should look.

Find p-value in Console

TYPE THIS IN THE SOURCE EDITOR

```
# H0: population mean score = 60
t.test(students$score, mu = 60)

# H0: probability of passing = 0.5
# Exactly 7 passed in 10 students
binom.test(7, 10, p = 0.5)

# P-value and confidence interval are in Console output.
```

H_0

Null hypothesis (e.g., $\mu = 60$).

H_1

Alternative hypothesis (e.g., $\mu \neq 60$).

p-VALUE

Probability, assuming H_0 and model are true, of data at least this extreme.

$\alpha = 0.05$

A prechosen decision threshold, not a guarantee.

1. **State H_0 and H_1 before testing**
   Decide whether your question calls for a one-sided or two-sided alternative.
2. **Run the appropriate command**
   Read the **p-value**, interval and test statistic in the Console.
3. **Write a careful conclusion**
   Compare p to your chosen threshold and explain what the evidence does or does not support.

REMEMBER

A small p-value is not the probability that H_0 is true, and statistical significance is not the same as practical importance.

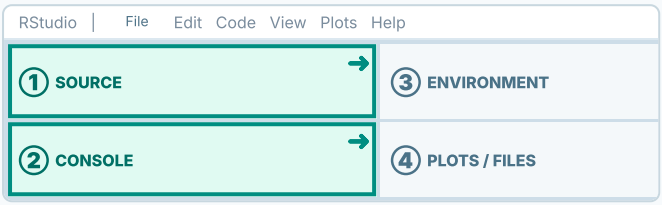
INFERENCE · WEEK 11

Two groups and chi-square tests

A comparison of group means differs from a test of association between categorical variables.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

COMPARE GROUPS ↓



Run t.test() or fisher.test().

Green box = where you should look.

Read test result in Console

TYPE THIS IN THE SOURCE EDITOR

```
# Independent-group comparison of scores
t.test(score ~ group, data = students)

# Create a pass/fail indicator
students$passed <- students$score >= 60

# Cross-tabulation: group by pass status
tab <- table(students$group, students$passed)
tab

# Chi-square: learn syntax, but check expected counts!
chisq.test(tab)$expected
# With tiny counts, prefer Fisher exact test
fisher.test(tab)
```

T-TEST

Compares **mean scores** between A and B under relevant assumptions.

FISHER / CHI-SQUARE

Tests whether two **categorical variables** are associated; sparse cells favor Fisher's exact test.

> Our ten-student table is deliberately tiny. Chi-square approximation conditions are not met when expected cell counts are too small. Inspect `chisq.test(tab)$expected` before using the chi-square p-value.

PAUSE & PRACTICE

When is Fisher's exact test preferable to chi-square approximation?

☐ DONE

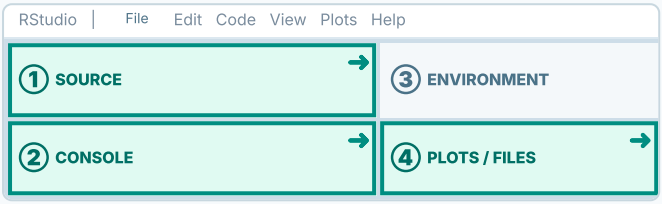
REGRESSION · WEEKS 12–13

Your first linear regression

Regression describes how a response variable changes with an explanatory variable in a fitted model.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

REGRESSION ↓



Fit lm() and draw the line.

Green box = where you should look.

Model summary left; plot right

TYPE THIS IN THE SOURCE EDITOR

```
model <- lm(score ~ hours, data = students)
summary(model)            # slope, intercept, p-value, R²
coef(model)               # fitted coefficients
confint(model)            # confidence intervals

plot(students$hours, students$score,
     xlab = "Study hours", ylab = "Exam score")
abline(model, col = "red", lwd = 2)

predict(model, newdata = data.frame(hours = 6))
```

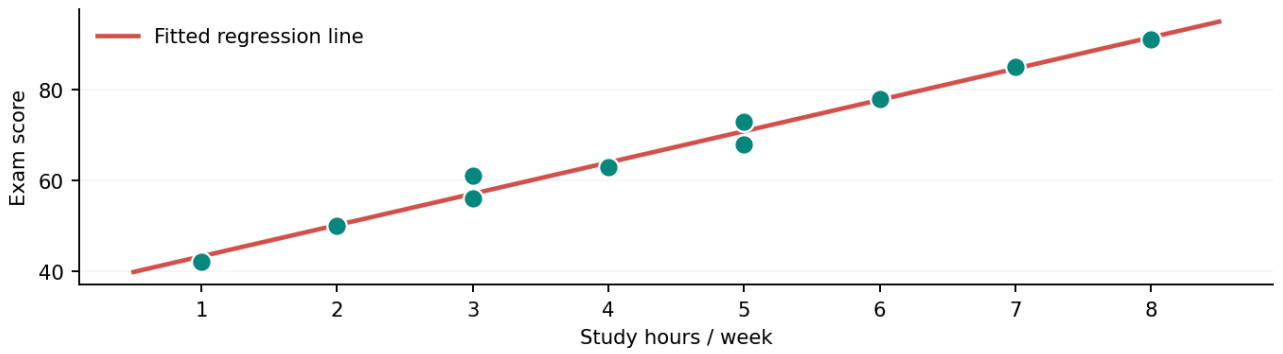


Figure 16. Regression line through the fictional student scores.

Fitted score = $\hat{\beta}_0 + \hat{\beta}_1 \times$ hours · Here $\hat{\beta}_1 \approx 6.90$ points per additional hour.

INTERPRETATION

On this fitted line, an additional study hour is **associated** with about 6.9 higher score points, not necessarily a causal effect. Inspect residuals and assumptions.

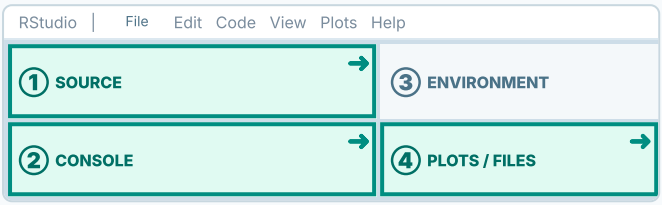
PRACTICAL WORKFLOW

One complete mini-project

Save a reproducible script. Reopening it should let you regenerate your core results.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

SAVE YOUR WORK ↓



Run code and export PNG.

Green box = where you should look.

Find score_vs_hours.png in Files

WHAT STUDENTS SUBMIT

A saved R script, one figure, and a short paragraph explaining the data, mean, relationship and limitations.

TYPE THIS IN THE SOURCE EDITOR

```
# STAT201 final mini-project (run after creating students)
stopifnot(nrow(students) == 10)
print(summary(students$score))
print(mean(students$score))
print(sd(students$score))
print(cor(students$hours, students$score))
model <- lm(score ~ hours, data = students)
print(summary(model))
png("score_vs_hours.png", width = 900, height = 600)
plot(students$hours, students$score, pch = 19,
     xlab = "Hours", ylab = "Score")
abline(model, col = "red", lwd = 2)
dev.off()  # close graphics file
```

1. **Save your source file**
   Press **Ctrl + S** to save the entire analysis in an **.R** file.

2. **Run your analysis**
   Choose the lines in order, press **Ctrl + Enter**, and read results.

3. **Find the exported figure**
   Click the RStudio **Files** tab and find **score_vs_hours.png** in the project folder.

WRITE THE CONCLUSION

"In these 10 fictional observations, the mean exam score is 66.7 points, and hours and scores show a strong positive association. These descriptive findings are not causal evidence about the larger population."

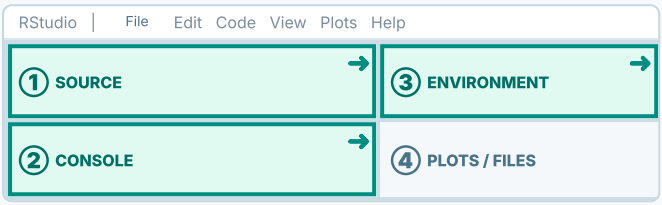
TROUBLESHOOTING

When R says "Error"

Errors are normal. Identify where you are typing, what exists, and what R expects.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

ERROR? CHECK ↓



Find object names and error text.

Green box = where you should look.

Error in Console; objects top right

`object 'scores' not found`	**You did not create the object yet.** Run the line `scores <- c(...)` first; names are case-sensitive.
`cannot open the connection`	**R cannot find the requested file.** Check `getwd()`, `list.files()` and your project folder.
`unexpected symbol`	**Your punctuation or quotation marks are wrong.** Look for a missing comma, unclosed quote, extra bracket or copied curly quote.
`could not find function`	**Function name is misspelled or package not loaded.** Use base R functions in this book; check spelling and capitalization.
`non-numeric argument`	**You tried arithmetic on a text field.** Use `str(students)` and ensure numeric variables are numeric.

FAST DIAGNOSIS

Type `ls()` to see objects, `getwd()` to see your folder, and `?mean` to open function help.

STUCK AT + IN CONSOLE?

R expects you to close a quote, parenthesis or bracket. Press **Esc** and correct the original line.

PAUSE & PRACTICE

Run getwd(), list.files(), and ls() to locate your current files and objects.

☐ DONE

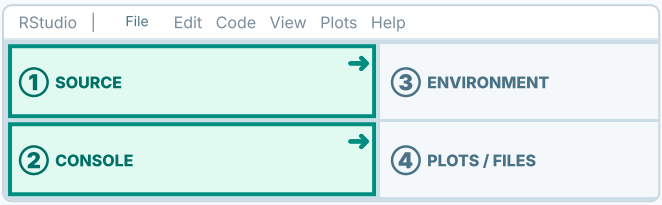
DESK REFERENCE

The one-page R command map

Use this page while completing STAT201 problem sets.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

QUICK REFERENCE ↓



Type the command in Source.

Green box = where you should look.

Check result in Console

GET STARTED

`x <- c(1,2,3)`	create vector
`data.frame(...)`	build data table
`read.csv("file.csv")`	import CSV
`View(students)`	inspect table

DESCRIPTIVE

`mean(x); median(x)`	center
`sd(x); var(x)`	dispersion
`IQR(x); quantile(x)`	quartiles
`table(group)`	frequencies

GRAPHS

`hist(x)`	histogram
`boxplot(x)`	boxplot
`plot(x, y)`	scatterplot
`abline(model)`	regression line

PROBABILITY

`dbinom(k,n,p)`	exact binomial mass
`pbinom(k,n,p)`	binomial cumulative
`pnorm(x,mu,sigma)`	normal cumulative
`qnorm(p,mu,sigma)`	normal quantile

INFERENCE

`t.test(x)`	t-based CI / test
`binom.test(k,n)`	binomial test
`fisher.test(tab)`	small-table association
`lm(y ~ x, data=d)`	regression

WORKFLOW

`Ctrl + Enter`	run source line
`Ctrl + S`	save source script
`getwd(); list.files()`	find files
`set.seed(123)`	repeat simulation

> On macOS replace Ctrl with Cmd for the shortcuts listed. The parameter names `mu` and `sigma` in this compact reference denote the numerical mean and standard deviation.

PAUSE & PRACTICE

Without looking: which commands find mean, binomial P(X=k), and correlation?

☐ DONE

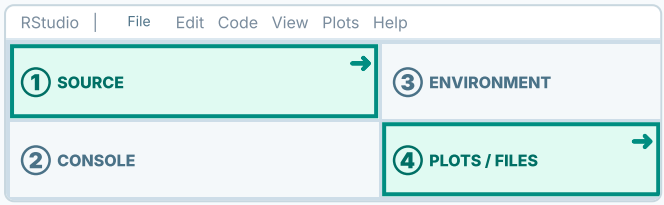
STUDENT ASSIGNMENT

Try it yourself · STAT201 practical 01

Complete this independently after following the guide. Put all answers in one saved .R file.

📍 FIND IT IN RSTUDIO · SAME 4-BOX MAP

STUDENT PROJECT ↓



Create and save assignment01.R.

Green box = where you should look.

Script shows in Files

Q1 Create a new RStudio project called **STAT201-YourName**. Save a script named **assignment01.R**.

Q2 Create a vector `rent <- c(200, 220, 250, 260, 450)`. Find its mean, median, sample SD and range.

Q3 Create a histogram of `rent`. Explain whether an extreme high rent affects the mean.

Q4 Using `dbinom()`, find the probability of exactly 2 successes in 4 trials if p = 0.5.

Q5 Using `pnorm()`, find P(X < 75) for X ~ Normal(70, 10^2).

Q6 Using the book's students data, plot study hours against scores and calculate their correlation.

Q7 Write 4–6 sentences describing what your results mean and what cannot be concluded.

SELF-CHECK

Rent mean = **276**; median = **250**. Binomial P(X=2) = **0.375**. Normal P(X<75) ≈ **0.6915**.

PAUSE & PRACTICE

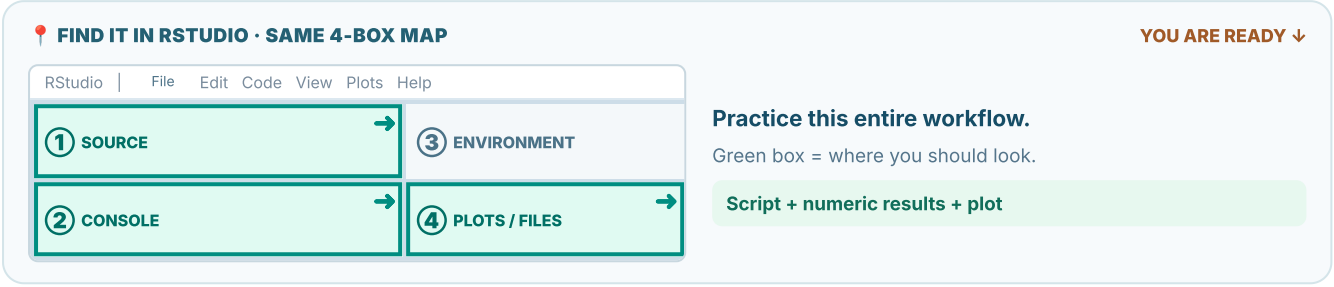
Save the completed assignment01.R file in your own project.

☐ DONE

END OF HANDBOOK

You can now start analysing data

Keep practicing. The next step is to combine statistical reasoning with reproducible code.



YOU NOW KNOW HOW TO...

Install R and RStudio, create a project, write and execute an R script, identify variables, import data, graph distributions and run introductory statistical analyses.

1. **Repeat Lesson 7 without looking**
   Create a script, define `scores <- c(42, 50, 56, 61)`, then run `mean(scores)` and interpret the answer.

2. **Revisit by week**
   Work with this book beside your TIU STAT201 lecture pages and problem sets.

3. **Keep one reproducible file**
   A good R assignment includes data creation/import, calculations, plots and interpretation.

Official links and further learning

R (CRAN) — official project
https://cran.r-project.org/

RStudio Desktop — free edition
https://posit.co/downloads/

RStudio interface and key panes
docs.posit.co/ide/user/ide/guide/ui/ui-panes.html

RStudio getting-started guide
docs.posit.co/ide/user/ide/get-started/

> The screen diagrams in this handbook are original simplified illustrations for navigation, not actual application captures. Labels and placement reflect the documented standard interface as of October 2026 but may vary by operating system, preferences or version.